

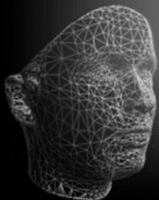
Advanced Topics in Computer Graphics II

Geometry Processing

Reconstruction from point clouds



January 9, 2025





Input: Sample of a surface F by an unordered set of points

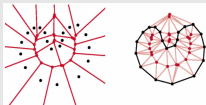
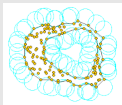
$$\mathcal{P} = \{p_1, \dots, p_n\} \quad , \quad p_i \in \mathbb{R}^3$$

Desired output: Reconstruct a triangle mesh out of this sample



Explicit

- Local surface connectivity estimation

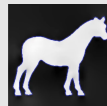


Left: Alpha Shapes (from CGAL User Manual), Right: Voronoi Filtering

- Point interpolation

Implicit

- Signed Distance Function (SDF) or Indicator Function estimation



Left: Signed Distance Function, Right: Indicator Function

- Mesh extraction



Explicit

- ▶ Alpha shapes (Edelsbrunner and Mücke, 1994 (3364 cit.))
- ▶ Mesh Zippering (Turk and Levoy, 1994 (2127 cit.))
- ▶ Voronoi filtering (Amenta and Bern, 1999 (1311 cit.))
- ▶ Ball pivoting algorithm (Bernardini et al., 1999(1855 cit.))
- ▶ Power Crust Amenta, Choi, et al., 2001 (1073 cit.))
- ▶ ...

Implicit

- ▶ Distance from tangent plane (Hoppe et al., 1992, (4617 cit.))
- ▶ Volumetric Range Image Processing (VRIP) (Curless and Levoy, 1996), (4191 cit.), KinectFusion, 2011, Newcombe et al., 2011, (5006 cit.))
- ▶ Poisson surface reconstruction (Kazhdan, Bolitho, et al., 2006, (3851 cit.))
- ▶ Global optimization for shape fitting (Lempitsky and Boykov, 2007, (181 cit.), Boykov, Veksler, et al., 2001, (10365))
- ▶ ...



Given: A sequence of depth maps

$$\mathcal{D}_1, \dots, \mathcal{D}_n$$

captured incrementally by a sensor in an on-line manner (only the current depth map is known)

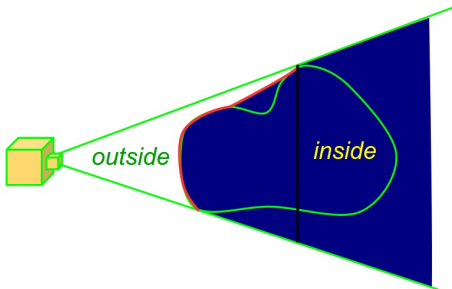
Problem: Reconstruct geometry using this scans¹

Idea: Exploit structure of depth images and take estimation errors and noise into account

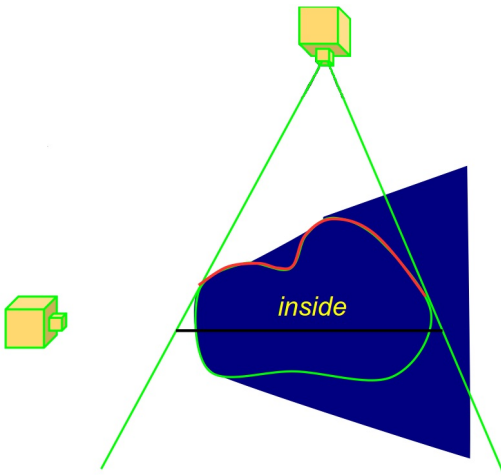
- ▶ Representation of range uncertainty
 - The data in range images typically have asymmetric error distributions with primary directions along sensor lines of sight
- ▶ Utilization of all range data
 - Including redundant observations of each object surface. If properly used, this redundancy can reduce sensor noise

¹B. Curless and M. Levoy. "A Volumetric Method for Building Complex Models from Range Images". In: *Proceedings of the 23rd Annual Conference on Computer Graphics and Interactive Techniques*. SIGGRAPH '96. ACM, 1996, pp. 303–312.

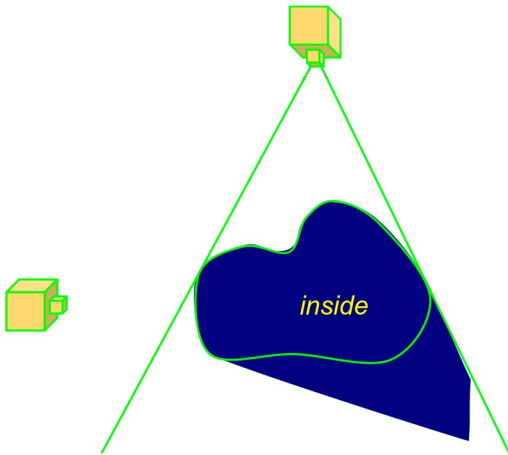
Idea:



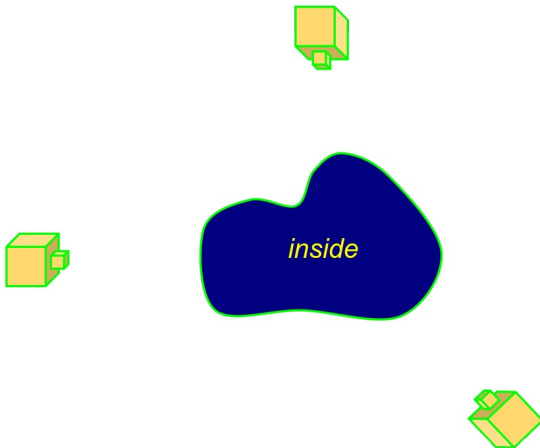
Idea:



Idea:



Idea:





For each depth image $\mathcal{D}_1, \dots, \mathcal{D}_n$ compute signed distance functions $d_1(\mathbf{x}), \dots, d_n(\mathbf{x})$:

- ▶ From every point $\mathbf{x}$ towards the next range surface along sensor line of sight
- ▶ Evaluation on discrete voxel grid

Implicit surface representation by distance functions $d_1(\mathbf{x}), \dots, d_n(\mathbf{x})$ weighted by functions $w_1(\mathbf{x}), \dots, w_n(\mathbf{x})$:

$$D(\mathbf{x}) = \frac{\sum_{i=1}^n w_i(\mathbf{x}) d_i(\mathbf{x})}{\sum_{i=1}^n w_i(\mathbf{x})}$$

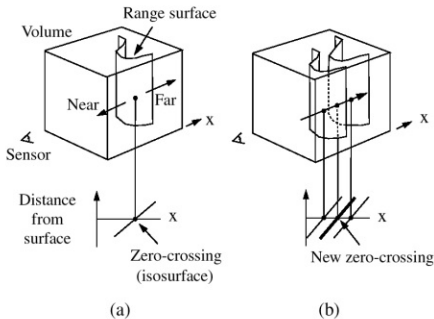
→ Extract implicit surface $D = 0$ using Marching Cubes algorithm



Example: Unweighted distance functions

$$\forall \mathbf{x} \in V: \forall i \in [1, n]: w_i(\mathbf{x}) = 1$$

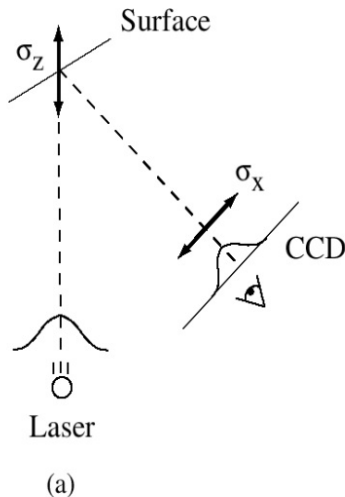
- ▶ One measurement: Zero crossing of function is point in range image
- ▶ Several measurements: Compute arithmetic mean





Consider 2D case:

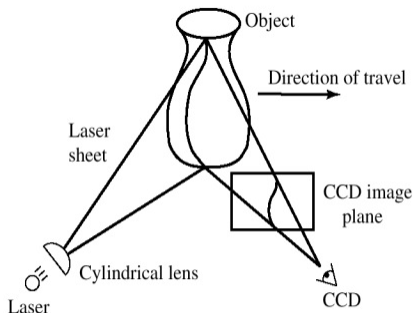
- ▶ Narrow laser beam illuminates the object surface
- ▶ Linear sensor images reflection
- ▶ Uncertainty σ_x in determining the center of the pulse results in range uncertainty σ_z along the laser's line of sight





Consider 3D case:

- ▶ Laser beam is spread into a sheet of light using a cylindrical lens
- ▶ Depth profile is computed from CCD observations of the reflected stripes
- ▶ Motions: Translation and rotation of the object or of the laser, respectively

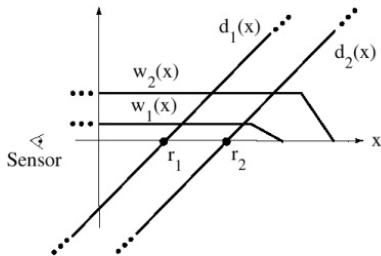


(b)

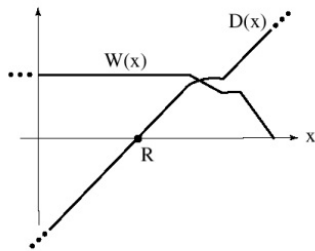


Weights are necessary to represent variations in certainty across the range surfaces

- ▶ Larger uncertainty when illumination is at grazing angles of the surface:
→ Make the weight depend on the dot product between each vertex normal and the viewing direction
- ▶ Range data at the boundaries of the mesh typically have larger uncertainty
→ Requires more down-weighting
- ▶ Additionally, choice of weights depends on the scanning technology in use



(a)

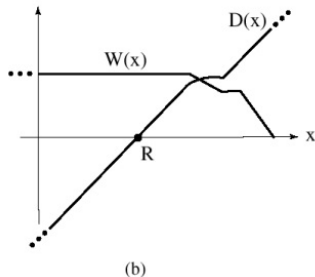
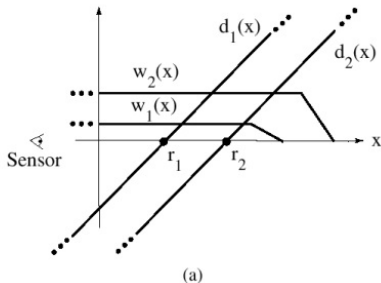


(b)



Properties of weighting functions:

- Force weights to fall off at a distance equal to half the maximum uncertainty interval of the range measurements
- Restricting the functions to the vicinity of the surface yields a more compact representation and reduces the computational expense of updating the volume





Problem: But how to update the volume without storing all previous measurements?

Lemma (Weighted Cumulative Moving Average)

Let $d_1(\mathbf{x}), \dots, d_{n+1}(\mathbf{x})$ be a sequence of measurements and $w_1(\mathbf{x}), \dots, w_{n+1}(\mathbf{x})$ be a sequence of weights. Then the weighted average

$$D_{n+1}(\mathbf{x}) = \frac{\sum_{i=1}^{n+1} w_i(\mathbf{x}) d_i(\mathbf{x})}{\sum_{i=1}^{n+1} w_i(\mathbf{x})}$$

can be computed on-line using the formula

$$D_{n+1}(\mathbf{x}) = \frac{W_n(\mathbf{x}) D_n(\mathbf{x}) + w_{n+1}(\mathbf{x}) d_{n+1}(\mathbf{x})}{W_n(\mathbf{x}) + w_{n+1}(\mathbf{x})}$$

$$W_{n+1}(\mathbf{x}) = W_n(\mathbf{x}) + w_{n+1}(\mathbf{x})$$

→ Only need to store 2 values to compute the weighted average!



Proof.

Consider

$$\begin{aligned}w_{n+1}(\mathbf{x}) d_{n+1}(\mathbf{x}) &= \sum_{i=1}^{n+1} w_i(\mathbf{x}) d_i(\mathbf{x}) - \sum_{i=1}^n w_i(\mathbf{x}) d_i(\mathbf{x}) \\&= \left(\sum_{i=1}^{n+1} w_i(\mathbf{x}) \right) D_{n+1}(\mathbf{x}) - \left(\sum_{i=1}^n w_i(\mathbf{x}) \right) D_n(\mathbf{x}) \\&= W_{n+1}(\mathbf{x}) D_{n+1}(\mathbf{x}) - W_n(\mathbf{x}) D_n(\mathbf{x})\end{aligned}$$

Rearranging the terms and using $W_{n+1}(\mathbf{x}) = W_n(\mathbf{x}) + w_{n+1}(\mathbf{x})$ leads to

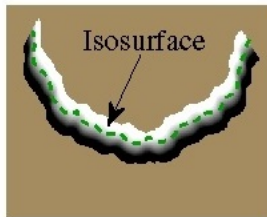
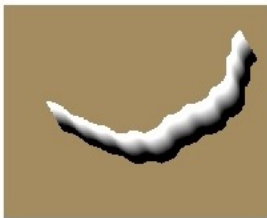
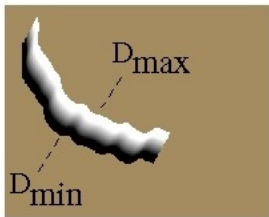
$$D_{n+1}(\mathbf{x}) = \frac{W_n(\mathbf{x}) D_n(\mathbf{x}) + w_{n+1}(\mathbf{x}) d_{n+1}(\mathbf{x})}{W_n(\mathbf{x}) + w_{n+1}(\mathbf{x})}$$





Observation: We only want to extract the isosurface, i.e. $D(\mathbf{x}) = 0$

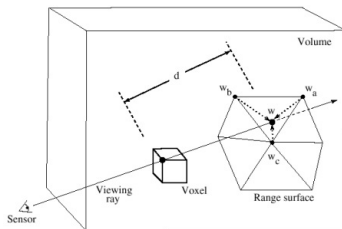
- ▶ Only store a small region around the measured surface
→ Define truncation-region δ
- ▶ Isosurface $D(\mathbf{x}) = 0$ invariant against scaling of the signed distance function
→ Define truncated signed distance function $tsdf(\mathbf{x}) = \frac{sdf(\mathbf{x})}{\delta}$ and cut off values above/below $D_{max} = 1, D_{min} = -1$
- ▶ Remaining volume is considered to be empty





Algorithm Volume Integration

- 1: Initialize all voxel weights to zero, i.e. $W = 0$
 - 2: **for** each voxel v in the volume **do**
 - 3: Cast a ray from the sensor through v to the surface
 - 4: Calculate the signed distance d from v to the measurement
 - 5: Calculate a weight w by interpolating weights of the measurements
 - 6: Update the stored distance D_i and weight W_i
 - 7: **end for**
-





Volume Integration

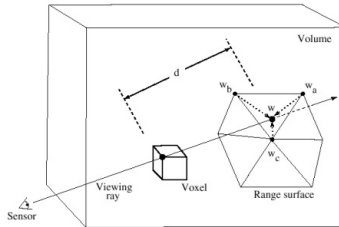


Theorem (Volume Integration)

The isosurface constructed by this algorithm is optimal in a least-squares sense

$$E(f) = \sum_{i=1}^n \int \int w(\hat{f}_i(x, y), f) \cdot d(\hat{f}_i(x, y), f)^2 dx dy$$

where f is the true surface and $\hat{f}_i(x, y)$ a measurement of f at position (x, y) in the i -th depth map.





Problem: Unseen portions of the surface will appear as holes in the reconstruction

- ▶ Aesthetically unsatisfying
- ▶ Can present a stumbling block to follow-on algorithms that require continuous meshes
- ▶ Example: rapid prototyping technologies such as stereo-lithography typically require a “watertight” model for constructing solid replica

Idea: Hole filling on reconstructed mesh

- ▶ Works well if regions of the mesh near each hole are nearly planar
- ▶ But holes in the meshes can be (and frequently are) highly non-planar

Instead: Use hole filling approach that operates on the volume

- ▶ Contains more information than the reconstructed mesh
- ▶ Classify all points in the volume into one of three states

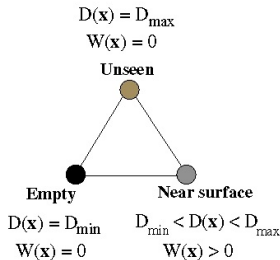
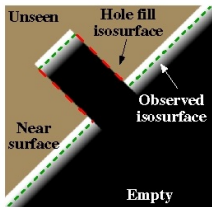
Unseen	$D(\mathbf{x}) = D_{max}$	$W(\mathbf{x}) = 0$
Empty	$D(\mathbf{x}) = D_{min}$	$W(\mathbf{x}) = 0$
Near surface	$D_{min} < D(\mathbf{x}) < D_{max}$	$W(\mathbf{x}) > 0$



Hole filling



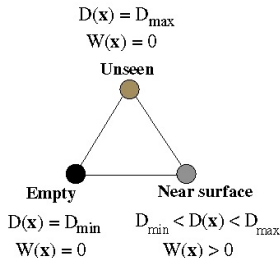
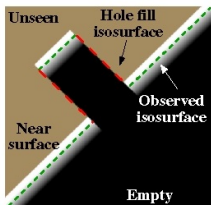
- ▶ Holes in the surface are indicated by frontiers between unseen regions and empty regions
- ▶ Surfaces placed at these frontiers offer a plausible way to plug these holes (red dashed line)
- ▶ Setting empty voxels to D_{min} and unseen voxels to D_{max} automatically defines an isosurface
- ▶ Signed distance (dotted green) and hole fill isosurfaces (dashed red) can be naturally connected where they meet, i.e. at the corners





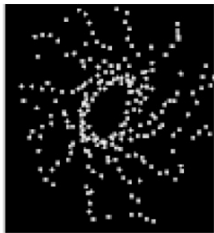
Algorithm Extended Volume Reconstruction

- 1: Initialize all voxels to state “unseen”, i.e. $D = D_{max}, W = 0$
- 2: Update the voxels near the surface as described previously (Volume Integration). These voxels take on continuous signed distance and weight values
- 3: Extract isosurface at zero-crossing of signed distance function. Additionally, extract a surface between regions seen to be empty and regions that remain unseen





Results:



(a)

Range data as
unstructured point
cloud



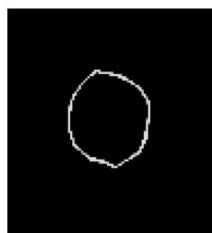
(b)

Range data as a
set of wireframe
tessellations



(c)

Reconstruction using
polygonal zippering
method (Turk et al. ,
Siggraph 1994)



(d)

Reconstruction
using Volume
Carving

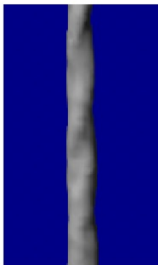


Results:



(e)

(e) Rendering of the zippered surface



(f)

(f) Rendering of the volumetrically generated surface



(g)

(g) Photo of the original drill bit which was painted white for scanning



Results:

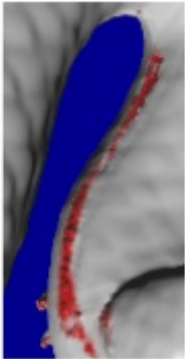


Reconstruction from 61 range images using space carving and hole filling (red).

- Wrong reconstruction at the frontier between unseen(brown) and empty (black) voxels causing extraneous triangles



Results:

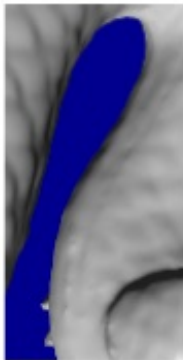


Reconstruction with 10 additional range images, space carving, hole filling and backdrop surfaces for improved carving

- Extraneous hole fill triangles nearly vanish



Results:



Additional filtering of hole filling portions

- Details in remaining mesh are preserved

The final model contains 1.8 million polygons and is watertight



Properties:

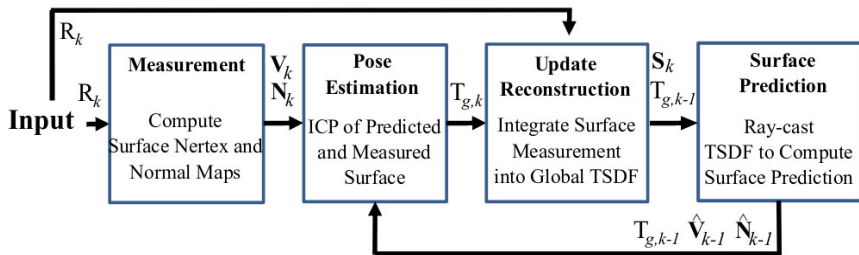
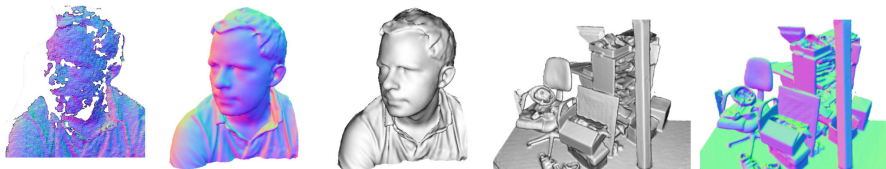
- ▶ Utilization of all range data
- ▶ Incremental and order independent updating
- ▶ Robustness towards sensor errors
- ▶ No restrictions regarding topology
- ▶ Ability to fill gaps in the reconstruction by carving space
- ▶ Resulting models have no holes, therefore suited for surface fitting, rapid prototyping, rendering

Limitations:

- ▶ Difficulties at sharp corners if no scan captures both surfaces meeting at the corner
→ Hole filling dampens this problem
- ▶ Problems with thin surfaces
→ Influence of observed surfaces extends behind their estimated positions for each range image and can interfere with distance functions originating from scans of the opposite side of a thin surface
- ▶ Similar problem with sharp corners
→ Artifacts such as thickening of surfaces or rounding of sharp corners



Idea: Real-time registration and volumetric integration on a GPU²³



²S. Izadi et al. "KinectFusion: Real-time 3D Reconstruction and Interaction Using a Moving Depth Camera". In: *Proc. of the ACM Symp. on User Interface Software and Technology*. 2011, pp. 559–568.

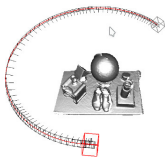
³R. A. Newcombe et al. "KinectFusion: Real-Time Dense Surface Mapping and Tracking". In: *Proc. of IEEE Int. Symp. on Mixed and Augmented Reality*. IEEE, 2011.



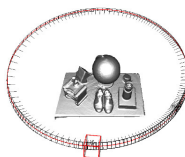
Pose Estimation: Hierarchical Point-to-Plane Iterative Closest Point (ICP) on 3 resolution levels



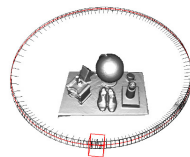
(a) Frame to frame tracking



(b) Partial loop



(c) Full loop

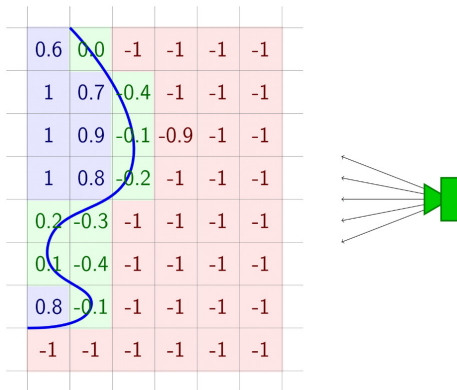


(d) M times duplicated loop

- ▶ Frame-to-Frame tracking accumulates drift resulting in a broken reconstruction
- ▶ Frame-to-Model tracking is significantly less susceptible to drift



Update Reconstruction: Volumetric Integration by Curless and Levoy (1996)



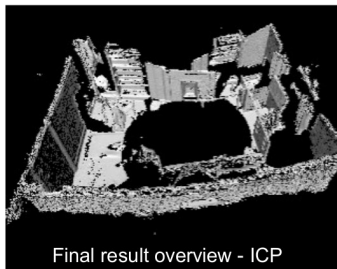
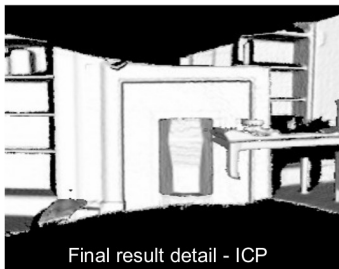
- Online update formula for the stored TSDF value:

$$D_{n+1}(\mathbf{x}) = \frac{W_n(\mathbf{x}) D_n(\mathbf{x}) + w_{n+1}(\mathbf{x}) d_{n+1}(\mathbf{x})}{W_n(\mathbf{x}) + w_{n+1}(\mathbf{x})}$$



Limitations:

- ▶ High memory consumption: $512 \times 512 \times 512$ voxel grid in GPU memory
→ Does not scale to larger scenes
- ▶ Frame-to-Model camera tracking
→ Requires strong geometric features in the scene
→ No explicit loop closure for globally consistent reconstruction



→ No recovery from tracking losses



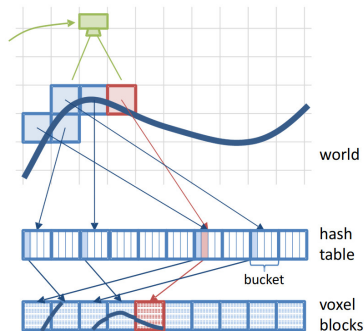
Idea: Sparse representation of the scene⁴

- ▶ Summarize $8^3 = 512$ voxel into a voxel block
- ▶ Only store voxel blocks inside the truncation region δ
- ▶ Manage unordered set of voxel blocks using spatial hashing

$$H(x, y, z) = (x \cdot p_1 \oplus y \cdot p_2 \oplus z \cdot p_3) \bmod n$$

where

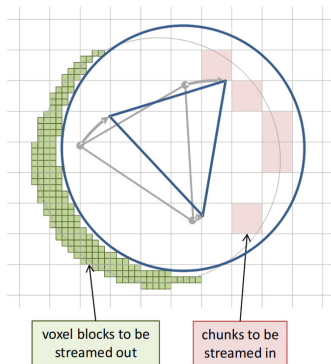
- ▶ $(x, y, z) \in \mathbb{Z}^3$ are the discrete voxel block coordinates
- ▶ $p_1 = 73856093$, $p_2 = 19349669$, $p_3 = 83492791$ are large prime numbers
- ▶ n is the number of buckets in the hash table
- ▶ $\oplus$ is the XOR operator



⁴M. Nießner et al. "Real-time 3D Reconstruction at Scale using Voxel Hashing". In: *ACM Transactions on Graphics (TOG)* (2013).



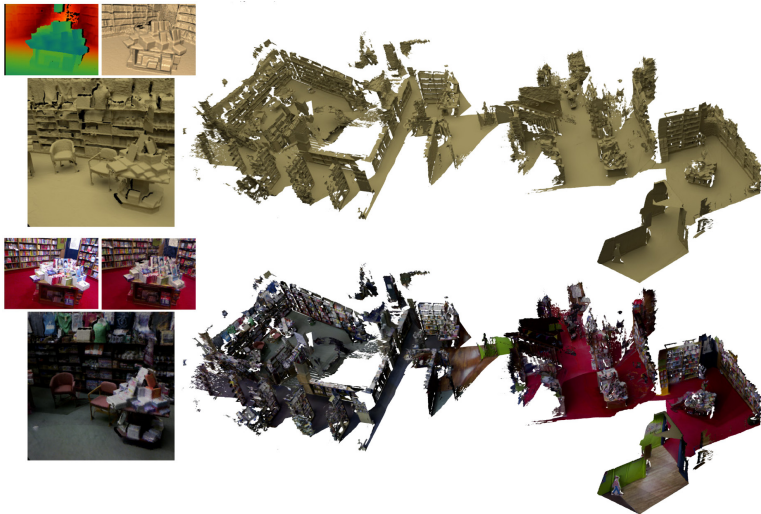
Idea: Sparse representation of the scene



- Stream voxel blocks between CPU and GPU memory
 - Allows to reconstruct much larger scenes
 - Voxel blocks are summarized as chunks on the CPU and managed using a simple list

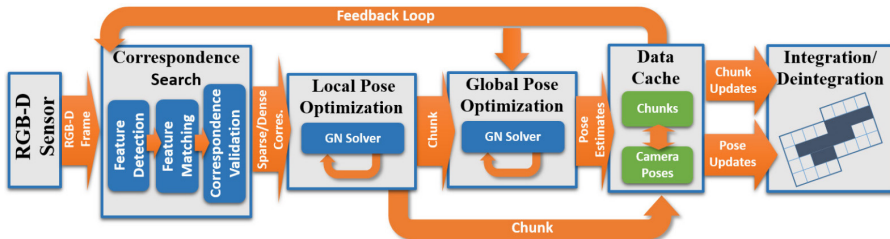


Results:





Idea: On-line global optimization of camera poses⁵



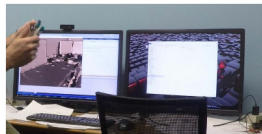
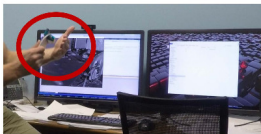
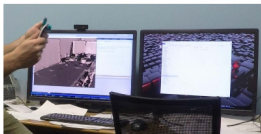
- ▶ Detect sparse SIFT features in RGB-D images
- ▶ Use chunks of 10 RGB-D images with 1 image overlap and perform local bundle adjustment of the SIFT features within the chunk
- ▶ Perform global bundle adjustment of the chunks for consistent camera pose tracking
- ▶ Incrementally integrate and deintegrate the chunks into the VoxelHashing-like scene representation

⁵A. Dai et al. "BundleFusion: Real-time Globally Consistent 3D Reconstruction using On-the-fly Surface Re-integration". In: *ACM Transactions on Graphics 2017 (TOG)* (2017).

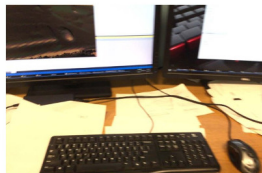
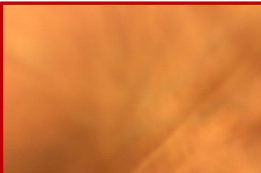
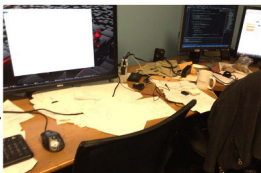


Results: Relocalization after tracking loss

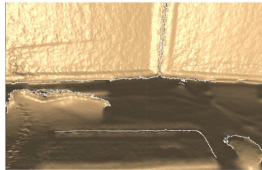
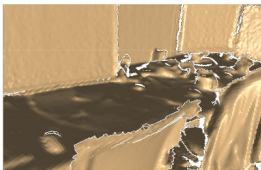
External View



Input Color

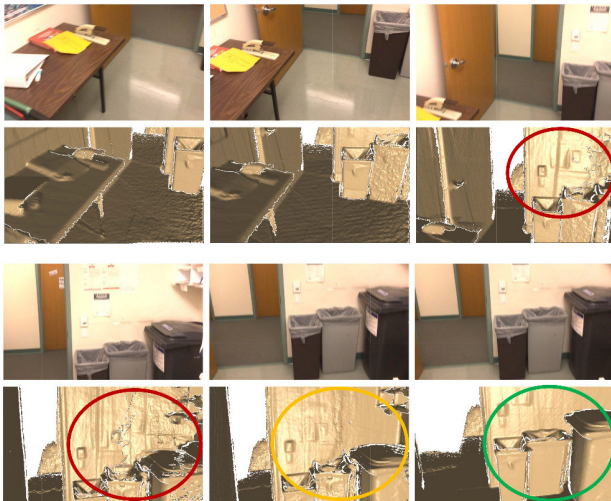


Reconstruction



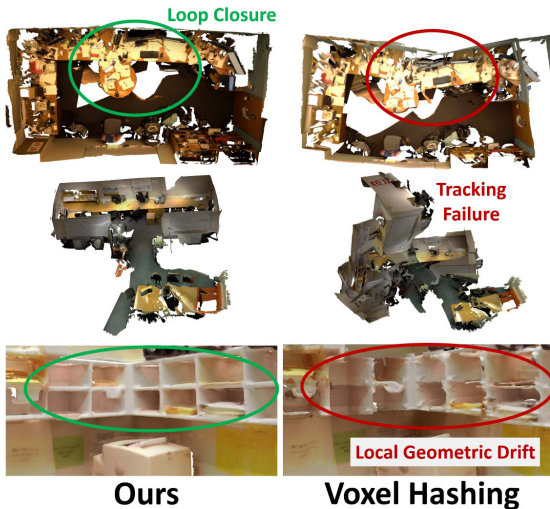


Results: Reintegration after loop closure





Results: Globally consistent 3D reconstructions

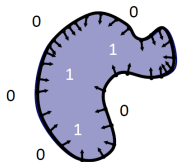




Poisson Surface Reconstruction



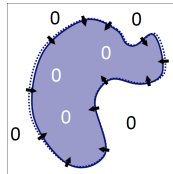
Idea: Solve for the **Indicator Function** $\chi_M(p) = \begin{cases} 1, & \text{if } p \in M \\ 0, & \text{if } p \notin M \end{cases}$ of the shape M



Indicator function χ_M .



Oriented input points (point-normal pairs).



Indicator gradient $\nabla\chi_M$.

Key observation:

Interpret the normal field $\mathbf{n}$ as **gradient** of "smoothed" indicator function $\hat{\chi}_M$:

$$\nabla \hat{\chi}_M = \mathbf{n}$$

Solve Poisson equation for $\hat{\chi}_M$:

$$\nabla^2 \hat{\chi}_M = \nabla \cdot \mathbf{n}$$



Task: Given m *gradient* data pairs $p_i, \mathbf{n}_i \in \mathbb{R}^3$, $\mathbf{n} = \hat{\chi}_M$ the smoothed normal field, find $\hat{\chi}_M : \mathbb{R}^3 \rightarrow \mathbb{R}$ such that

$$\int_{\Omega} \|\nabla \hat{\chi}_M(\mathbf{u}) - \mathbf{n}(\mathbf{u})\|_2^2 d\mathbf{u}$$

is minimized.

Note, that this is the variational problem discussed in Lecture 8:

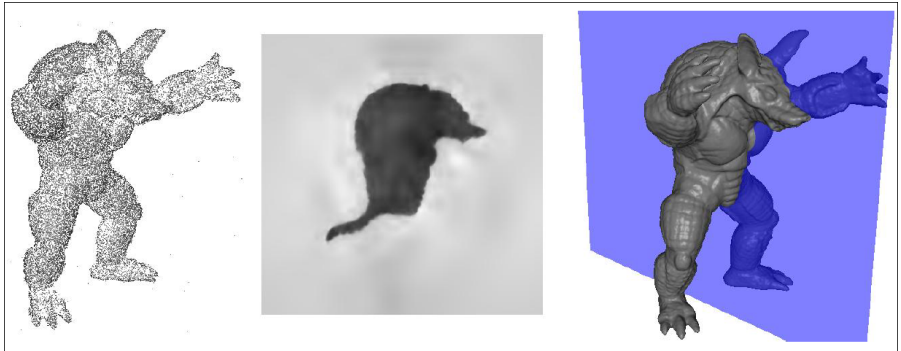
$$\begin{aligned} &\text{minimize}_{\hat{\chi}_M} \int_{\Omega} \|\nabla \hat{\chi}_M(\mathbf{u}) - \mathbf{n}(\mathbf{u})\|_2^2 d\mathbf{u} \\ &\text{subject to } \hat{\chi}_M(\mathbf{u}) = f_0(\mathbf{u}) \quad \forall \mathbf{u} \in \partial\Omega. \end{aligned}$$

Minimizing leads to the **Poisson equation**:

$$\begin{aligned} \Delta \hat{\chi}_M(\mathbf{u}) &= g(\mathbf{u}) & \mathbf{u} &\in \Omega \\ \hat{\chi}_M(\mathbf{u}) &= f_0(\mathbf{u}) & \mathbf{u} &\in \partial\Omega, \end{aligned}$$

where $g(\mathbf{u}) = \operatorname{div} \mathbf{n}(\mathbf{u}) = \nabla \cdot \mathbf{n}(\mathbf{u})$, $\mathbf{u} \in \Omega$ and f_0 is a function describing the boundary conditions.

Example:



Points from scans of the “Armadillo Man” model (left), The Poisson surface reconstruction (right), and a visualization of the indicator function (middle) along a plane through the 3D volume. (from Kazhdan, 2006)



A Galerkin formulation is used to transform this problem into a finite-dimensional system.

1. Represent $\hat{\chi}_M$ as

$$\hat{\chi}_M = \sum_{j=i}^k \alpha_i B_i(\|x - c_i\|), \quad i = 1, \dots, N$$

where c_i are the centers of the cells of an octree which is generated from the input points and the B_i are basis functions associated with each node of the octree.

$$\langle \Delta \hat{\chi}_M, B_i \rangle_{[0,1]^3} = \langle \nabla \cdot \mathbf{n}, B_i \rangle_{[0,1]^3}, \quad 1 \leq i \leq N$$

2. Solve a linear system $\mathbf{L}\alpha = \mathbf{b}$ for the coefficients α_j , where

$$L_{ij} = \langle \nabla B_i, \nabla B_j \rangle_{[0,1]^3} \text{ and } b_i = \langle \mathbf{n}, \nabla B_i \rangle_{[0,1]^3}$$

In the original paper⁶ the n -th convolution of a box filter with itself was chosen which was centered at the center of the j -th cell:

$$B_j(x, y, z) = (B(x) \cdot B(y) \cdot B(z))^{*n} \text{ with } B(t) = \begin{cases} 1, & |t| < 0.5 \\ 0, & \text{otherwise} \end{cases}$$

⁶M. Kazhdan, M. Bolitho, et al. "Poisson surface reconstruction". In: *Proceedings of the fourth Eurographics symposium on Geometry processing*. Vol. 7. 2006.



Given the points:

- ▶ Set octree
- ▶ Compute vector field
- ▶ Compute indicator function
- ▶ Extract iso-surface

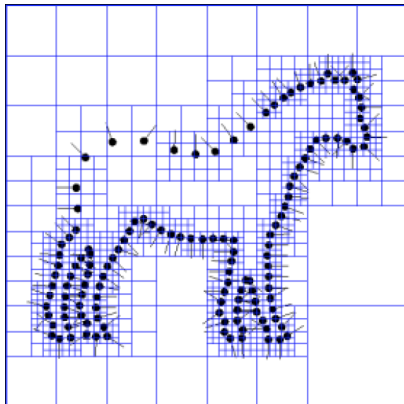


Image from C. Gotsman & M. Kazhdan, 2008



Given the points:

- ▶ Set octree
- ▶ Compute vector field
 - ▶ Chose function space
 - ▶ Splat the samples
- ▶ Compute indicator function
- ▶ Extract iso-surface

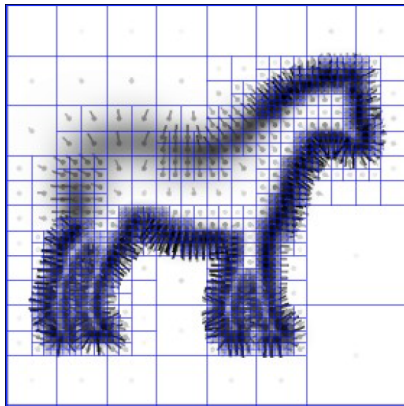


Image from C. Gotsman & M. Kazhdan, 2008



Given the points:

- ▶ Set octree
- ▶ Compute vector field
- ▶ Compute indicator function
 - ▶ Compute divergence
 - ▶ Solve Poisson equation
- ▶ Extract iso-surface

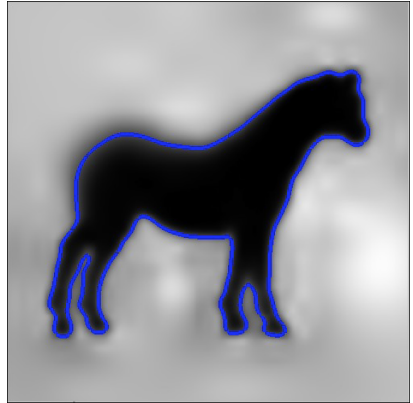


Image from C. Gotsman & M. Kazhdan, 2008



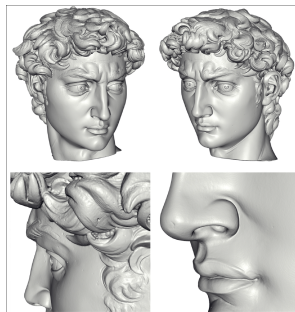
Given the points:

- ▶ Set octree
- ▶ Compute vector field
- ▶ Compute indicator function
- ▶ Extract iso-surface





Left: VRIP (Curless & Levoy), Right: Poisson Reconstruction



Reconstruction of the head of Michelangelo's David (maximum tree depth of 11). Even fine features in the model such as the inset iris, the drill marks in the hair, the chip on the eyelid, and the creases around the nose and mouth are visible.

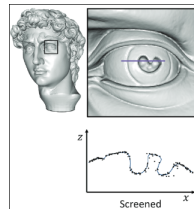
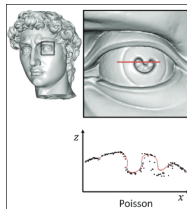


Improvements



Screened Poisson Reconstruction, Kazhdan and Hoppe, 2013

- Higher fidelity - at same triangle count
- Faster - solver time is linear



Reconstruction of the head of Michelangelo's David (maximum tree depth of 10). The 2D visualizations show cross-section curves from the reconstructions overlaid with adjacent validation points.

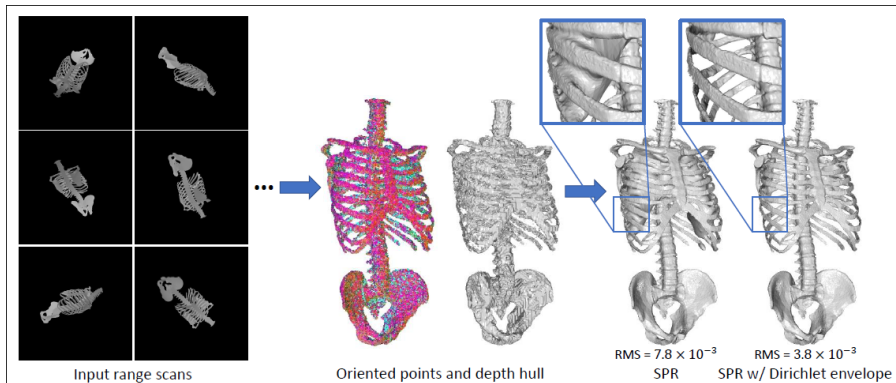


Improvements



Poisson Reconstruction with envelope constraints, Kazhdan, Chuang, et al., 2020

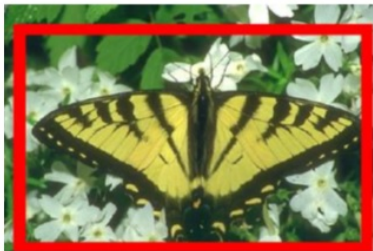
- Incorporates the depth hull as Dirichlet constraints within the Poisson formulation
- Prevents the emergence of extraneous surfaces, resulting in a more accurate model





Excursus: Image Segmentation





Problem: Segment the image $\mathcal{I}$ in foreground and background

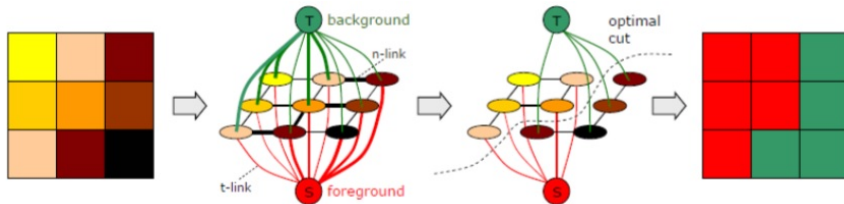
Idea: Pose segmentation problem as an energy (or cost) minimization⁷

$$E(L) = \underbrace{\sum_{p \in \mathcal{I}} D_p(L_p)}_{\text{Data term}} + \underbrace{\sum_{(p,q) \in \mathcal{I}} V_{(p,q)}(L_p, L_q)}_{\text{Smoothness term}}$$

⁷C. Rother et al. "Grabcut: Interactive foreground extraction using iterated graph cuts". In: *ACM Transactions on Graphics (TOG)* 23.3 (2004), pp. 309–314.



Idea: State the energy minimization as a graph cut problem



Define a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$:

► Nodes and Edges:

$$\mathcal{V} = \bigcup_{i=1}^n \{x_i\} \cup \underbrace{\{S\}}_{\text{source}} \cup \underbrace{\{T\}}_{\text{sink}}, \quad \mathcal{E} = \underbrace{\{N\}}_{\text{n-links}} \cup \bigcup_{i=1}^n \{(x_i, S), (x_i, T)\}$$

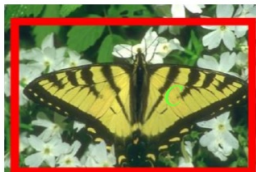
► Edge weights represents energy

- n-links (neighborhood links): smoothness term
- t-links: (terminal links) data term



Observation: Segmentation can be regarded as an implicit function in a volume

- Can define interior (foreground) Ω_1 and exterior (background) Ω_0



$$\Omega_1 \equiv \{p \mid L_p = 1\} = \text{int}(C)$$

$$\Omega_0 \equiv \{p \mid L_p = 0\} = \text{ext}(C)$$

Idea: Use this observation to define data and smoothness terms⁸

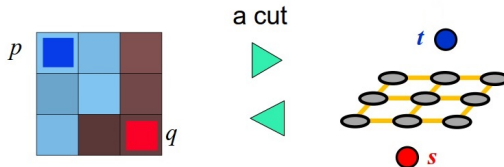
$$E(L) = \sum_{p \in \mathcal{I}} D_p(L_p) + \sum_{(p,q) \in \mathcal{I}} w_{pq} \cdot \delta(L_p \neq L_q)$$

→ Smoothness term: Only add a cost if two neighboring pixel have different segmentation states

⁸Y. Boykov and M.-P. Jolly. "Interactive graph cuts for optimal boundary & region segmentation of objects in N-D images". In: *Computer Vision, 2001. ICCV 2001. Proceedings. Eighth IEEE International Conference on*. Vol. 1. 2001, 105–112 vol.1.



Question: How to define the data terms?



Hard constrains

Soft constrains

$$p \in \Omega_0$$

$$p \in \Omega_1$$

$$D_p(L_p = 1) = \infty \quad D_p(L_p = 1) = 0$$

$$D_p(L_p = 0) = 0 \quad D_p(L_p = 0) = \infty$$

$$D_p(L_p = 1) = -\log \Pr(I_p \mid L_p = 1)$$

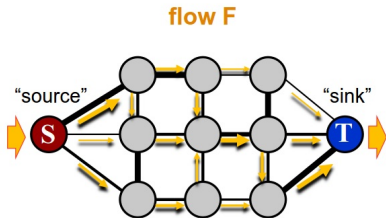
$$D_p(L_p = 0) = -\log \Pr(I_p \mid L_p = 0)$$

Either foreground or background

Uncertainty using color distributions for foreground and background

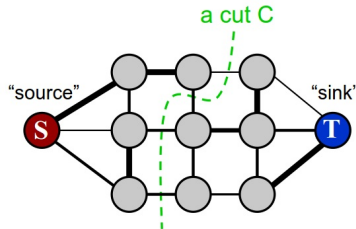


Max-Flow



- ▶ Each edge is a “pipe”
- ▶ Find the largest flow F of “water” that can be sent from the “source” to the “sink” along the pipes
- ▶ Edge weights give the pipe’s capacity

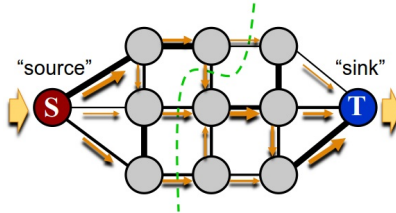
Min-Cut



- ▶ Find the cheapest way to cut the edges so that the “source” is completely separated from the “sink”
- ▶ Edge weights now represent cutting “costs”



$$\text{Max-Flow} = \text{Min-Cut}^9$$



- ▶ Maximum flow saturates the edges along the minimum cut
- ▶ Problem reduction!
- ▶ Ford and Fulkerson (1962) gave first polynomial time algorithm for globally optimal solution

⁹L. Ford and D. R. Fulkerson. *Flows in networks*. Vol. 1962. Princeton University Press, 1962.



Shape Fitting





Shape reconstruction



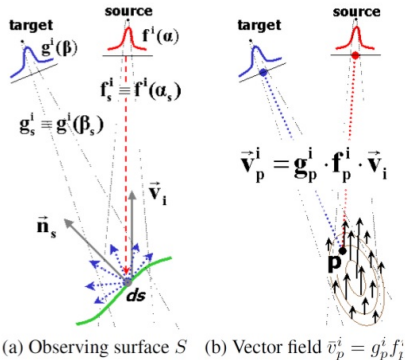
Given: (Sparse, noisy) measurements $\mathcal{P}$ with (weak) estimated normals $\mathcal{N}$:

$$\mathcal{P} = \{p_1, \dots, p_n\} \quad , \quad \mathcal{N} = \{n_1, \dots, n_n\}$$

Question: How to measure how well a surface is fitted to noisy data?

Idea: Model the problem probabilistically with probability functions ρ_i and a vector field $\vec{V}$ on $\mathbb{R}^3$!

- Use sparse vector field $\{v_1, \dots, v_n\}$ defined by normals ...
- ... and smooth it out by a blur filter to get a semi-dense field $\vec{V}$



$$\vec{V} = \{\vec{v}_p = \sum_i \rho_i(p) v_i \mid \forall p \in \mathbb{R}^3\}$$



$$\vec{V} = \{\vec{v}_p = \sum_i \rho_i(p) \mathbf{v}_i \mid \forall p \in \mathbb{R}^3\}$$

Solution: Maximize flux through $\vec{V}$ to get a segmentation/shape fit¹⁰

- Data term (fit to measurements)

$$- \int_S \langle \vec{v}_s \mid \vec{n}_s \rangle dS$$

- Smoothness term (surface area minimal)

$$\int_S \lambda dS$$

- Occupancy term (optional)

$$- \int_{interior(\vec{V})} O(\vec{v}_p) dV$$

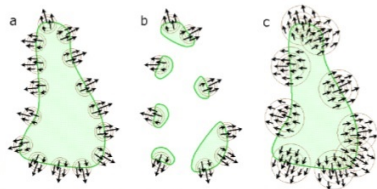


Figure 6. Surface fitting with different sampling density and priors. a) Optimization of the data-fit term in combination with a membrane prior produces consistent reconstruction for densely sampled points. b) The same fails when the sampling density is low. c) Widening the support of each point (ad-hoc “low curvature” prior) allows to obtain correct reconstruction.

¹⁰V. Lempitsky and Y. Boykov. “Global optimization for shape fitting”. In: *Computer Vision and Pattern Recognition, 2007. CVPR’07. IEEE Conference on*. IEEE. 2007, pp. 1–8.



So the energy to minimize is defined as

$$E(S) = - \int_S \langle \bar{\mathbf{v}}_s | \bar{\mathbf{n}}_s \rangle dS + \int_S \lambda dS - \int_{\text{interior}(\vec{V})} O(\bar{\mathbf{v}}_p) dV$$

Problem: Data term is defined over the surface for which we want to optimize

Theorem (Divergence theorem)

Let $\text{interior}(\vec{V})$ be a solid region inside a vector field $\vec{V}$ in $\mathbb{R}^3$ and S be the boundary surface of this region with positive orientation to the exterior. Furthermore, let the components of $\vec{V}$ have continuous first-order partial derivatives. Then

$$\int_S \langle \vec{V} | \mathbf{n}(S) \rangle dS = \int_{\text{interior}(\vec{V})} \text{div } \vec{V} dV$$

This theorem implies the following:

- ▶ The flux through the surface S at each position with $\langle \vec{V} | \mathbf{n}(S) \rangle dS = \langle \bar{\mathbf{v}}_s | \bar{\mathbf{n}}_s \rangle dS$ is equal to ...
- ▶ ... the divergence of the vector field with $\text{div } \vec{V} dV = \text{div } \bar{\mathbf{v}}_p dV$

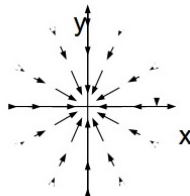
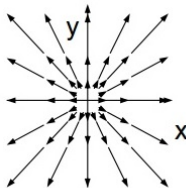


The divergence is defined as

$$\operatorname{div} \mathbf{v} := \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} =: \langle \nabla | \mathbf{v} \rangle$$

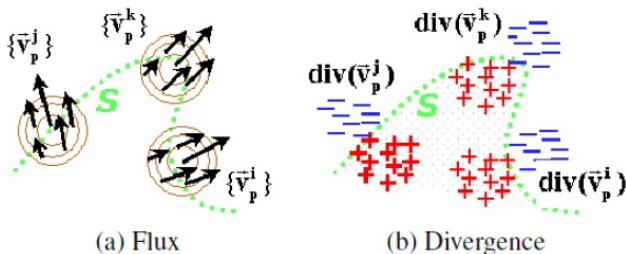
Properties of divergence:

- ▶ $\operatorname{div} \mathbf{v}$ is a scalar
- ▶ $\operatorname{div} \mathbf{v}(\mathbf{x}_0) > 0$: $\mathbf{v}$ has a source in $\mathbf{x}_0$
- ▶ $\operatorname{div} \mathbf{v}(\mathbf{x}_0) < 0$: $\mathbf{v}$ has a sink in $\mathbf{x}_0$
- ▶ $\operatorname{div} \mathbf{v}(\mathbf{x}_0) = 0$: $\mathbf{v}$ is source-free in $\mathbf{x}_0$
- ▶ Describes flow into/out of a region





Interpretation:



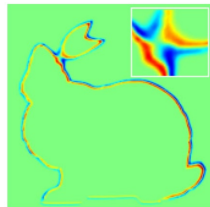
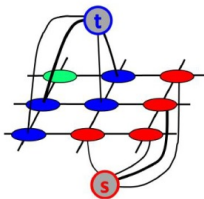
(a) Maximization of flux through vector field can be seen as maximization of the number of data points contained by a surface while allowing measurement uncertainty.

(b) This is equivalent to maximizing divergence of vector field inside surface interior. Points with non-zero (positive or negative) values of scalar function $\text{div}(\vec{v}_p)$ are indicated by + or -.



Now the energy can be rewritten using the Divergence theorem:

$$E(S) = \int_S \lambda dS - \int_{\text{interior}(\vec{V})} (\text{div } \bar{v}_p + O(\bar{v}_p)) dV$$



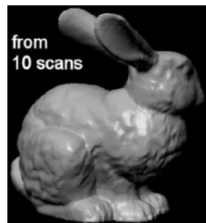
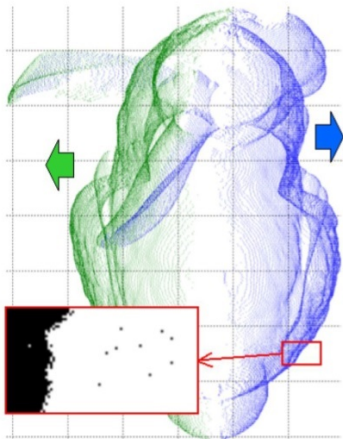
Convert energy into graph:

- ▶ Neighboring nodes are connected via n-links representing regularization cost¹¹
- ▶ The nodes are also connected to terminals via t-links representing volumetric potentials

¹¹Y. Boykov and V. Kolmogorov. "Computing geodesics and minimal surfaces via graph cuts". In: *Computer Vision, 2003. Proceedings. Ninth IEEE International Conference on*. IEEE, 2003, pp. 26–33.

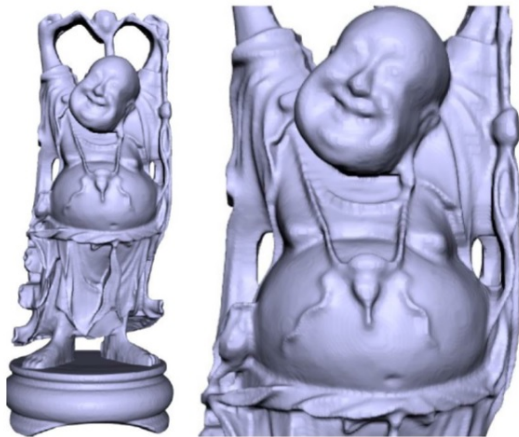


Results:





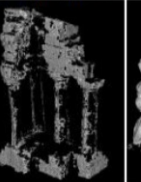
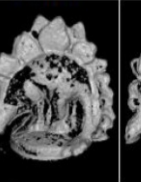
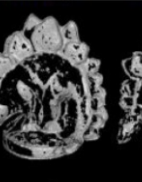
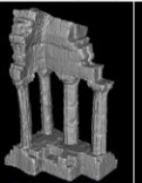
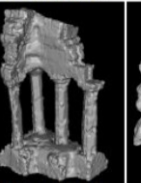
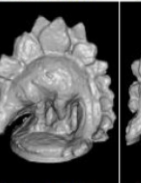
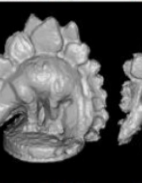
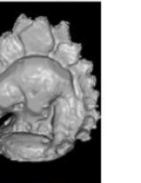
Results:



Buddha



Results: Fitting to incomplete meshes

Temple Full	Temple Ring	Temple Sparse	Dino Full	Dino Ring	Dino Sparse
					
0.42mm 98.0%	0.61mm 86.2%	0.87mm 56.6%	0.56mm 80.0%	0.46mm 57.8%	0.56mm 26.0%
					
0.42mm 99.7%	0.54mm 99.7%	0.95mm 93.2%	0.52mm 99.0%	0.64mm 95.9%	1.22mm 88.8%



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